

## EMPORIA, KS, USA

WMO: 724556

Lat: 38.329N

Lon: 96.195W

Elev: 365

StdP: 97.02

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 13989

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -15.1      | -12.4 | -20.1                         | 0.7 | -13.1 | -17.6 | 0.8 | -11.0 | 12.8                  | 7.8  | 11.9 | 5.8  | 4.1                   | 330  | 0.579 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.7                   | 37.1            | 23.7 | 34.9 | 23.9 | 33.0 | 23.6 | 25.7                | 32.7 | 25.1 | 32.1 | 24.5 | 31.0 | 5.9                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 23.8                            | 19.5 | 29.3 | 23.0 | 18.5 | 28.5 | 22.5 | 18.0 | 28.0 | 81.3          | 32.9 | 78.5 | 32.1 | 75.8 | 31.1 | 28.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |     |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-----|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |     |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB  | WB  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |     |
| (4) 12.3          | 11.1 | 9.7 | -19.5                      | 38.9 | 2.7                | 2.1 | -21.4                                              | 40.4 | -23.0      | 41.6 | -24.5      | 42.8 | -26.5      | 44.3 |     |     |
| (5)               |      |     | -19.9                      | 26.9 | 2.6                | 0.6 | -21.7                                              | 27.4 | -23.3      | 27.7 | -24.7      | 28.0 | -26.6      | 28.4 | (4) | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 13.4   | -0.2  | 1.8   | 7.5   | 13.3  | 18.6  | 23.6  | 26.2  | 25.4  | 21.0  | 14.2  | 7.4   | 1.2   |  |  |
|                                                                        | DBStd       | 10.58  | 6.33  | 6.35  | 6.22  | 5.19  | 4.34  | 3.30  | 3.15  | 3.32  | 4.19  | 5.16  | 5.57  | 5.90  |  |  |
|                                                                        | HDD10.0     | 1113   | 319   | 237   | 121   | 26    | 1     | 0     | 0     | 0     | 19    | 113   | 277   |       |  |  |
|                                                                        | HDD18.3     | 2623   | 576   | 462   | 337   | 164   | 52    | 3     | 0     | 1     | 20    | 147   | 329   | 532   |  |  |
|                                                                        | CDD10.0     | 2350   | 2     | 8     | 44    | 124   | 267   | 407   | 503   | 477   | 329   | 149   | 34    | 4     |  |  |
|                                                                        | CDD18.3     | 818    | 0     | 0     | 2     | 13    | 59    | 161   | 245   | 219   | 99    | 19    | 1     | 0     |  |  |
|                                                                        | CDH23.3     | 8501   | 0     | 1     | 28    | 147   | 519   | 1579  | 2737  | 2326  | 973   | 179   | 12    | 0     |  |  |
|                                                                        | CDH26.7     | 3797   | 0     | 0     | 3     | 32    | 144   | 636   | 1386  | 1153  | 393   | 48    | 1     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 4.7    | 4.7   | 4.9   | 5.4   | 5.7   | 4.9   | 4.6   | 4.0   | 3.8   | 4.2   | 4.6   | 4.9   | 4.7   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 903    | 22    | 34    | 57    | 86    | 126   | 122   | 115   | 96    | 89    | 68    | 53    | 34    |  |  |
|                                                                        | PrecMax     | 1254   | 71    | 81    | 128   | 214   | 239   | 243   | 383   | 204   | 193   | 175   | 254   | 82    |  |  |
|                                                                        | PrecMin     | 527    | 2     | 0     | 12    | 18    | 22    | 30    | 11    | 15    | 32    | 3     | 0     | 4     |  |  |
|                                                                        | PrecStd     | 188    | 17    | 24    | 34    | 48    | 56    | 61    | 84    | 50    | 52    | 45    | 54    | 25    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 18.6  | 21.9  | 26.4  | 29.9  | 32.1  | 36.4  | 39.1  | 39.0  | 36.4  | 30.9  | 25.0  | 19.1  |  |  |
|                                                                        |             | MCWB   | 11.1  | 12.1  | 16.7  | 18.9  | 21.9  | 24.3  | 23.4  | 22.9  | 22.8  | 20.7  | 15.6  | 13.3  |  |  |
|                                                                        | 2%          | DB     | 15.2  | 18.0  | 23.2  | 27.2  | 29.9  | 33.3  | 37.3  | 37.1  | 32.8  | 27.7  | 22.0  | 15.6  |  |  |
|                                                                        |             | MCWB   | 8.5   | 11.0  | 14.6  | 17.6  | 21.3  | 24.0  | 24.1  | 23.4  | 22.5  | 18.4  | 14.5  | 10.4  |  |  |
|                                                                        | 5%          | DB     | 12.2  | 14.9  | 20.6  | 24.8  | 28.2  | 31.9  | 35.3  | 34.7  | 31.0  | 25.0  | 18.9  | 12.5  |  |  |
|                                                                        |             | MCWB   | 7.1   | 9.6   | 13.3  | 16.1  | 20.7  | 23.9  | 24.2  | 23.7  | 22.1  | 17.3  | 12.4  | 8.1   |  |  |
|                                                                        | 10%         | DB     | 9.0   | 12.0  | 17.8  | 22.4  | 26.4  | 30.5  | 33.1  | 32.6  | 28.8  | 22.6  | 16.2  | 9.9   |  |  |
|                                                                        |             | MCWB   | 5.0   | 7.5   | 12.1  | 15.6  | 19.8  | 23.1  | 24.2  | 23.3  | 21.0  | 16.6  | 11.5  | 6.4   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 13.4  | 14.9  | 18.7  | 20.5  | 24.0  | 26.2  | 26.6  | 26.3  | 24.8  | 22.5  | 17.7  | 15.3  |  |  |
|                                                                        |             | MCDB   | 15.9  | 18.4  | 24.5  | 27.2  | 29.5  | 33.0  | 33.8  | 32.5  | 31.2  | 27.6  | 21.7  | 17.8  |  |  |
|                                                                        | 2%          | WB     | 9.8   | 12.3  | 16.7  | 19.1  | 22.9  | 25.3  | 25.8  | 25.4  | 23.8  | 20.9  | 15.4  | 11.5  |  |  |
|                                                                        |             | MCDB   | 14.4  | 16.3  | 20.3  | 24.3  | 28.1  | 31.6  | 33.1  | 32.4  | 30.1  | 25.0  | 19.3  | 13.9  |  |  |
|                                                                        | 5%          | WB     | 7.3   | 10.0  | 15.0  | 17.9  | 21.8  | 24.5  | 25.2  | 24.7  | 22.9  | 19.3  | 13.9  | 8.6   |  |  |
|                                                                        |             | MCDB   | 11.4  | 14.8  | 19.1  | 22.6  | 26.5  | 30.4  | 32.6  | 31.7  | 29.1  | 22.9  | 17.7  | 11.8  |  |  |
|                                                                        | 10%         | WB     | 5.2   | 7.2   | 12.6  | 16.5  | 20.7  | 23.7  | 24.7  | 23.9  | 22.0  | 17.6  | 12.0  | 6.5   |  |  |
|                                                                        |             | MCDB   | 9.4   | 11.6  | 17.2  | 21.5  | 25.2  | 29.2  | 31.5  | 30.6  | 27.3  | 21.5  | 15.6  | 9.7   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.1  | 11.4  | 12.4  | 12.3  | 11.2  | 11.1  | 11.7  | 12.1  | 12.4  | 12.4  | 12.0  | 10.7  |  |  |
|                                                                        |             | MCDBR  | 15.7  | 16.6  | 16.5  | 15.7  | 12.8  | 12.3  | 14.2  | 14.7  | 14.1  | 14.9  | 15.7  | 14.8  |  |  |
|                                                                        |             | MCWBR  | 10.0  | 10.5  | 9.4   | 8.5   | 6.5   | 5.3   | 4.5   | 4.7   | 5.6   | 7.4   | 9.1   | 10.0  |  |  |
|                                                                        |             | MCDBR  | 14.9  | 15.3  | 13.2  | 13.3  | 11.3  | 11.4  | 12.0  | 12.4  | 12.2  | 11.9  | 13.1  | 13.5  |  |  |
|                                                                        | 5% WB       | MCWBR  | 10.2  | 10.5  | 8.6   | 8.2   | 6.6   | 5.6   | 4.8   | 5.0   | 5.5   | 7.2   | 8.7   | 10.1  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.286 | 0.300 | 0.333 | 0.377 | 0.414 | 0.426 | 0.429 | 0.430 | 0.382 | 0.339 | 0.307 | 0.284 |  |  |
|                                                                        | taud        |        | 2.488 | 2.461 | 2.414 | 2.320 | 2.280 | 2.269 | 2.303 | 2.282 | 2.375 | 2.480 | 2.500 | 2.509 |  |  |
|                                                                        | Ebn at Noon |        | 903   | 933   | 929   | 901   | 869   | 855   | 850   | 840   | 864   | 876   | 869   | 881   |  |  |
|                                                                        | Edh at Noon |        | 78    | 92    | 107   | 125   | 133   | 134   | 129   | 129   | 111   | 90    | 77    | 72    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.39  | 3.13  | 4.06  | 5.16  | 5.83  | 6.65  | 6.55  | 5.87  | 4.87  | 3.58  | 2.59  | 2.05  |  |  |
|                                                                        | RadStd      |        | 0.23  | 0.40  | 0.30  | 0.33  | 0.50  | 0.45  | 0.45  | 0.32  | 0.34  | 0.41  | 0.23  | 0.19  |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | -0.90   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (27 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page